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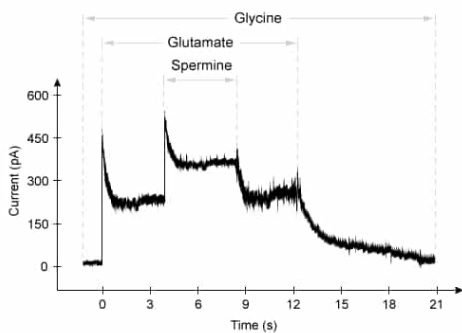


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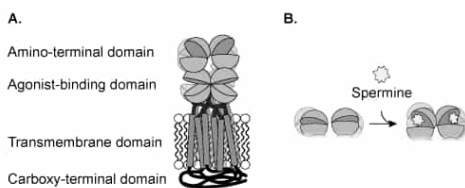


Figure 2 Cartoon structure of NMDAR with agonists and with and without spermine. (A) The amino-terminal and agonist-binding domains are extracellular, and the carboxy-terminal domain is intracellular. The 12-transmembrane helix transmembrane domain lies in between. (B) A close-up of the conformational changes seen in the amino-terminal domain upon spermine binding.

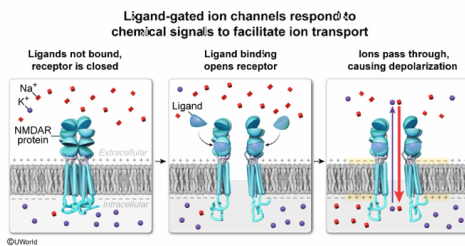
The NMDAR can best be described as a:

- ☐ A. voltage-gated ion channel. [10%]
- ☐ B. G-protein-coupled receptor. [7%]
- ☒ C. ligand-gated ion channel. [79%]
- ☐ D. receptor tyrosine kinase. [2%]

Correct

79% answered correctly

Explanation:



Ligand-gated ion channels are **membrane proteins** that facilitate the movement of ions across the membrane. These proteins are closed at rest, with the passage of ions blocked by a closed gate. However, the presence of **agonists**, which bind to proteins as ligands in order to activate them, induces conformational changes that result in the opening of the gate and the passage of ions through the proteins themselves.

NMDARs bind glutamate and glycine as ligands, resulting in the opening of the gates to its sodium- and potassium-permeable channels. Sodium rushes down its **electrochemical gradient**, whereas the flow of potassium out of the cell is slowed by the **negative membrane potential**. The influx of sodium causes local **depolarization** of the membrane which, if sufficient in magnitude, crosses the **threshold** to activate voltage-gated sodium channels and **propagate the electric signal**. Therefore, the NMDAR is a ligand-gated ion channel that facilitates excitatory signaling.

(Choice A) Voltage-gated channels use changes in membrane potential as the primary signal to open their ion channels, whereas NMDARs depend on the binding of glutamate and glycine (ligands).

(Choice B) **G-protein–coupled receptors (GPCRs)**, also known as 7-transmembrane receptors, bind extracellular ligands to activate the intracellular G-protein that is coupled to it. The activated G-protein then affects production of various **second messengers**. The NMDAR does not have a 7-transmembrane structure, nor is there any mention in the passage of G-proteins or second messengers.

(Choice D) Receptor tyrosine kinases are a type of receptor enzyme, which bind an extracellular ligand to activate an intracellular catalytic domain. No mention is made of NMDARs catalyzing any chemical reactions.

Educational objective:

Ligand-gated ion channels bind chemical molecules, resulting in their activation. Once activated, these proteins open an ion-permeable pore that allows for the passage of ions across a membrane, which can affect the membrane potential.

Time spent: 0 sec

Subject:

Biochemistry

QID: 403212

System:

Amino Acids and Proteins

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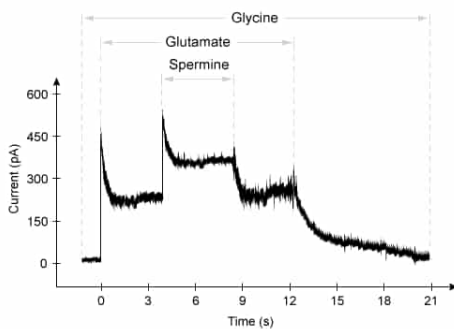


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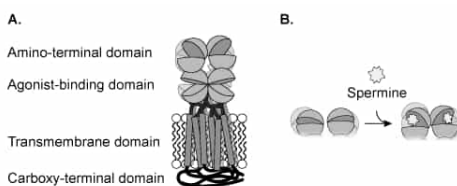


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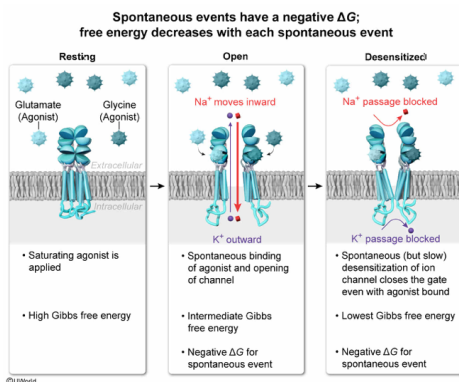
Under saturating agonist conditions, which functional state describes the NMDAR conformation that has the lowest Gibbs free energy?

- ✓ ☒ A. Bound to agonists and impermeant to ions [37%]
☐ B. Bound to agonists and permeant to ions [41%]
☐ C. Unbound to agonists and permeant to ions [7%]
☐ D. Unbound to agonists and impermeant to ions [13%]

Correct

37% answered correctly

Explanation:



The change in **Gibbs free energy (G)** for a process is the difference between the free energy of the product and of the reactant (ie, $\Delta G = G_{\text{products}} - G_{\text{reactants}}$). The sign of ΔG is a useful measure of the spontaneity of a reaction or event. Reactions that do not require energy input (spontaneous reactions) have a negative ΔG ($\Delta G < 0$). Given the formula for ΔG , it follows that the products of spontaneous reactions have a lower free energy than the reactants, as the subtraction of a larger value from a smaller value yields a negative result. Thus, the lowest free energy is found at the end of a spontaneous event or chain of events.

The passage indicates that NMDAR binds glutamate and glycine without external energy input, demonstrating that bound proteins have a lower free energy than unbound proteins. The passage further states that proteins undergo desensitization under prolonged agonist application. Though desensitization is slow (due to a **high energy barrier**), this process also does not require added energy and is spontaneous. Taken together, the desensitized form of NMDAR, which is impermeant to ions but still bound to agonists, has the lowest free energy.

(Choice B) Though the NMDAR *initially* opens spontaneously and allows ion passage when bound to a ligand, with prolonged application the receptors eventually enter a desensitized and impermeant state that has even *lower* free energy than the open state.

(Choices C and D) Under saturating conditions, proteins will spontaneously bind their ligands, including the agonists that bind NMDAR. Therefore, binding decreases the free energy of the proteins.

Educational objective:

All physical processes are governed by the laws of thermodynamics, from chemical reactions to protein conformational changes. Spontaneous reactions that do not require energy input are characterized by a negative ΔG and a lowering of their Gibbs free energy.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403213

System:
Amino Acids and Proteins

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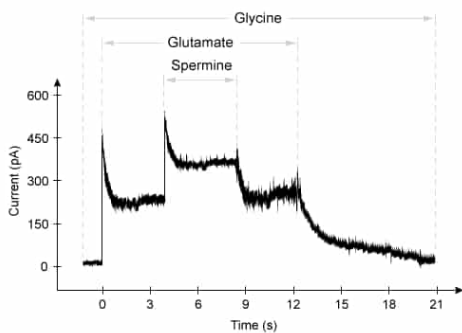


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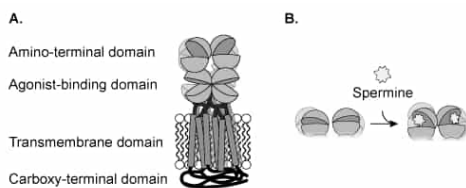


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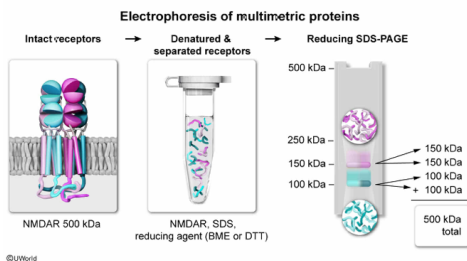
Which of the following band patterns could be produced if NMDARs were subjected to reducing SDS-PAGE?

- ✓ ☒ A. One band at 100 kDa and one band at 150 kDa [33%]
☐ B. One band at 500 kDa [13%]
☐ C. One band at 250 kDa [20%]
☐ D. One band at 200 kDa and one band at 300 kDa [32%]

Correct

34% answered correctly

Explanation:



Polyacrylamide Gel Electrophoresis (PAGE) is an analytic technique to separate proteins by their mobility through a highly crosslinked gel matrix. **Sodium dodecyl sulfate (SDS)** is a detergent that both **denatures** proteins and imparts a negative charge to minimize the effects of the protein's native charge and shape (conformation). Reducing agents (typically either **β-mercaptoethanol [BME]** or **dithiothreitol [DTT]**) ensure that any covalent **disulfide bonds** that stabilize tertiary or quaternary **structure** are also disrupted. This leaves size (molecular weight) as the major relevant factor affecting migration, with smaller proteins migrating faster toward the bottom of the gel.

The passage states that NMDAR is a heterotetramer, meaning it consists of four polypeptide subunits whose **quaternary structure** is mostly held together by noncovalent interactions. SDS and the reducing agents disrupt these interactions, allowing for each polypeptide subunit to travel through the gel individually rather than as an intact protein. Because it is a *heterotetramer* (rather than a homotetramer), the subunits are not identical and are likely different sizes.

However, because NMDAR is a dimer of identical dimers, each subunit in one dimer must be identical to one of the subunits in the other dimer. Therefore, two total bands are expected (one for each set of equivalent monomers). The four monomeric subunits must have a combined mass of 500 kDa. Multiple mass combinations could meet these criteria, but of the given options only Choice A meets all requirements:

$$100 \text{ kDa} + 150 \text{ kDa} + 100 \text{ kDa} + 150 \text{ kDa} = 500 \text{ kDa}$$

(Choice B) A single band at 500 kDa implies that the protein stays intact instead of being separated into monomers by SDS and the reducing agents.

(Choices C and D) The NMDAR is a tetramer and must have four subunits. With a single band at 250 kDa, the 500 kDa total will be reached with only two subunits. Similarly, bands at 200 kDa and 300 kDa will reach the 500 kDa total with only two subunits (one from each band).

Educational objective:

SDS and reducing agents denature proteins. When analyzed by reducing SDS-PAGE, multimeric proteins will lose their quaternary structure and run as individual monomers.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403214
System:
Amino Acids and Proteins

The *N*-methyl-D-aspartate receptor (NMDAR) is a 500 kDa heterotetrameric protein that is one of the main mediators of excitatory signaling between neurons in the mammalian brain. Arranged as a dimer of identical heterodimers, one subunit in each dimer binds glutamate molecules and the other binds glycine molecules that are released by the pre-synaptic neuron. Upon binding to both of these agonists, the NMDAR subsequently allows for the passage of sodium and potassium ions across the postsynaptic membrane by passing them through its transmembrane domain. However, in the overly prolonged presence of agonists, the receptor undergoes a process known as desensitization, wherein it adopts a new form that does not allow for the passage of ions despite continued agonist binding.

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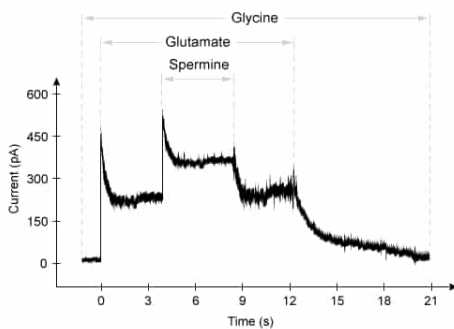


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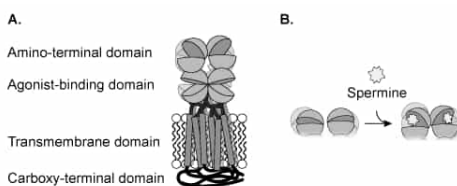


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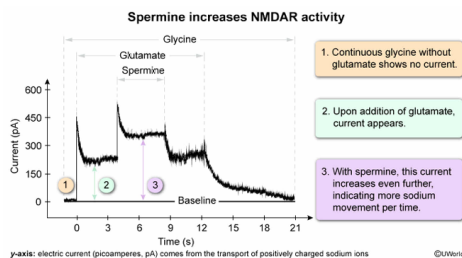
Based on the passage, what is the best description of the effect of spermine binding to the NMDAR?

- ☐ A. Spermine is a co-agonist, necessary for NMDAR function. [8%]
- ☐ B. Spermine binding changes the equilibrium membrane potential for sodium. [21%]
- ☐ C. Spermine increases the Hill coefficient to above 1. [16%]
- ☒ D. Spermine increases the rate of sodium transfer. [54%]

Correct

54% answered correctly

Explanation:



Positive regulators are molecules that increase the activity or function of a protein. While the protein being regulated can still act without the positive regulator, the interaction of the regulator with the protein enhances the preexisting function.

Figure 1 shows the effects of glycine, glutamate, and spermine on NMDAR activity as measured by electric current. With only glycine applied, there is no activity. Upon addition of the second co-agonist glutamate, the electric current signal increases, showing function if and only if both co-agonists are present. The additional application of spermine causes this current to **increase even further**.

Current is the movement of charge per unit time. The passage indicates that NMDARs allow for the movement of charged sodium and potassium ions, indicating that the current is coming from **cation movement** across the membrane. Therefore, an increase in current reflects an increase in the rate of ion movement, suggesting that when spermine binds the receptor, sodium ions are transferred across the membrane faster than when spermine is not bound.

(Choice A) Figure 1 shows that the NMDAR has activity even prior to application of spermine; thus, spermine is *not* required for activity.

(Choice B) The equilibrium membrane potential for an ion is the membrane potential at which there is no net ion flow across the membrane, because the electric and chemical gradients are balanced. Although the binding of spermine may alter the *rate* at which this potential is achieved, it will not directly affect ion concentrations, the chemical gradient, or the value of the equilibrium potential itself.

(Choice C) The **Hill coefficient** is a measure of cooperativity of ligand binding, which is a property that affects how easily proteins become saturated with ligands. While spermine may or may not have an effect on cooperativity, the experiments described in the passage cannot be used to draw any conclusions because the concentrations of ligands used are already saturating, regardless of spermine's inclusion.

Educational objective:

Certain molecules may affect, or modulate, a protein's function. These positive modulators are not required for protein function but simply increase the magnitude of the protein's activity.

Time spent: 0 sec
Subject:
 Biochemistry

QID: 403215
System:
 Amino Acids and Proteins

The *N*-methyl-D-aspartate receptor (NMDAR) is a 500 kDa heterotetrameric protein that is one of the main mediators of excitatory signaling between neurons in the mammalian brain. Arranged as a dimer of identical heterodimers, one subunit in each dimer binds glutamate molecules and the other binds glycine molecules that are released by the pre-synaptic neuron. Upon binding to both of these agonists, the NMDAR subsequently allows for the passage of sodium and potassium ions across the postsynaptic membrane by passing them through its transmembrane domain. However, in the overly prolonged presence of agonists, the receptor undergoes a process known as desensitization, wherein it adopts a new form that does not allow for the passage of ions despite continued agonist binding.

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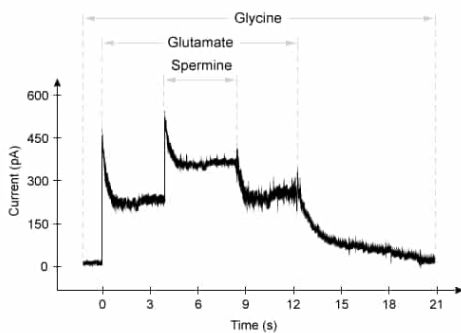


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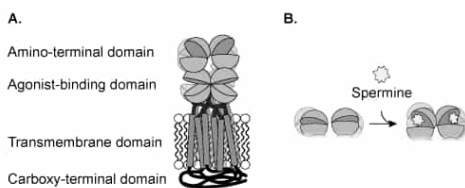


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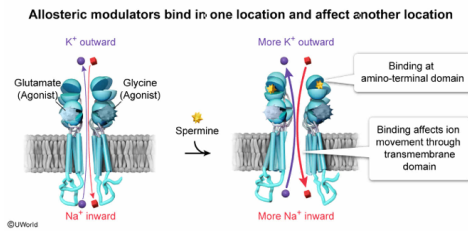
Which of the following best describes the most likely mechanism for spermine's effect on NMDAR function?

- ☐ A. The positive charges of the amino groups alter the membrane potential and thus the electrochemical gradients of sodium and potassium. [22%]
- ☒ B. Spermine alters the conformation and/or conformational stability of the transmembrane domain of the NMDAR. [72%]
- ☐ C. Spermine recruits kinases to phosphorylate the NMDAR. [3%]
- ☐ D. Spermine signaling induces a change in rate of translation of NMDAR mRNA. [1%]

Correct

72% answered correctly

Explanation:



The function of a protein is dictated by its shape (also known as its **structure** or **conformation**). Any alteration to that shape, whether via **mutations**, **post-translational modifications**, or small molecules that physically interact with the protein, will likely have some effect on protein function. This is the basis for regulation of protein function.

In many cases, direct changes at one part of the protein have measurable effects at a different location. In the passage, spermine binds to the extracellular amino-terminal domain, with effects seen on the rate of cation movement through the transmembrane domain. Though **physically distant** from each other, the effects of binding at the amino-terminal domain must alter the transmembrane domain to allow more ion transport. The conformational changes are carried through the various levels of protein structure. This effect—non-covalent binding at one location having an effect at a separate location—is known as **allostery**.

(Choice A) The electric potential across a membrane is caused by the *separation* of charges, not merely by the presence of charged species. Though a polyamine will be positively charged at neutral pH, unless it is separated from its counter anions it will have no effect on the electric or chemical gradients of sodium or potassium.

(Choice C) Though post-translational modifications like phosphorylation can affect protein function, there is no evidence presented that would suggest phosphorylation of the NMDAR by spermine.

(Choice D) Changes in protein expression levels are slow, long-term means of affecting activity in a cell; however, Figure 1 shows that the effects of spermine are activated and inactivated within milliseconds, suggesting that it does not work by altering any protein expression levels.

Educational objective:

Allosteric modulators are small molecules that reversibly bind to proteins to affect protein function. Allosteric modulators bind to a site distant from the site of action but affect activity through conformational changes that are transmitted through the levels of protein structure.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403216

System:
Amino Acids and Proteins

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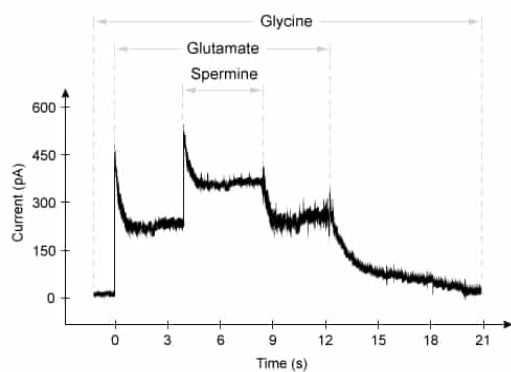


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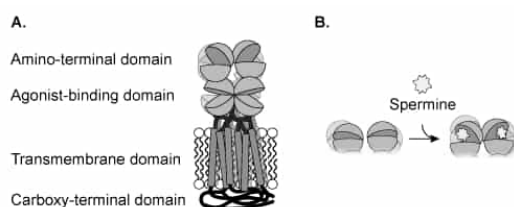


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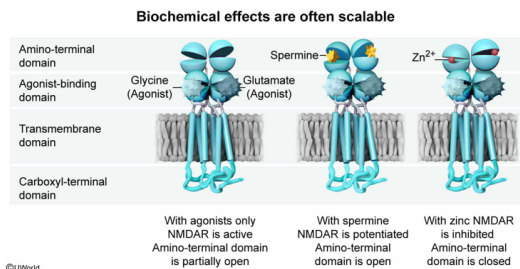
The divalent cation Zn^{2+} has an inhibitory effect on NMDAR receptors and is known to bind to the same domain as spermine. Researchers hypothesize that Zn^{2+} and spermine act via a similar mechanism at the amino-terminal domain to exert their effects. If Zn^{2+} is applied to NMDARs, which of the following observations would best support this hypothesis?

- ☐ A. Decreased electric resistance through open NMDAR channels [15%]
- ✓ ☒ B. A more closed-clamshell conformation of the amino-terminal domain [73%]
- ☐ C. Decreased trafficking of NMDAR to the plasma membrane [5%]
- ☐ D. Increased permeability to potassium but not to sodium [6%]

Correct

73% answered correctly

Explanation:



Although some all-or-none mechanisms exist (eg, [action potential](#)), many biochemical processes are scalable, with different degrees of activity corresponding to different strengths of the stimulus. According to the passage, spermine changes the conformation of the amino terminal domain from partially open to fully open, and spermine also has an [allosteric](#) effect that increases NMDAR function.

From these data, conclusions can be drawn and a **hypothesis** made regarding the amino-terminal domain conformation and protein function. If opening the amino-terminal domain increases NMDAR function, then closing the domain likely has an inhibitory effect. Zinc ions (Zn^{2+}) bind to the same domain as spermine yet have an inhibitory effect; therefore, the most likely observation to support this hypothesis is that zinc will inhibit NMDAR function by closing the amino-terminal domain.

(Choice A) A decreased electric resistance means that current flows more easily. This would be seen as a stimulating effect on NMDAR function rather than as inhibitory.

(Choices C and D) While both these choices will lead to decreased NMDAR activity, there is no evidence given in the passage that points to amino-terminal domain modulators having any effect on [protein trafficking](#) to the plasma membrane or ion selectivity.

Educational objective:

Allosteric modulators cause changes in protein function by inducing conformational changes. Modulators that affect conformation in similar or in opposite ways are likely to show similar or opposite effects on protein function, respectively.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403217

System:
Amino Acids and Proteins